Find and Fix: Illegal Cancellation and Lost Solutions Answer Key

Precalculus

Quick Answers

1. (a) all solutions = 0, 60, 180, 300 degrees, (b) named error = —
2. (a) all solutions = 30, 150, 210, 330 degrees, (b) named error = —
3. (a) all solutions = 60, 90, 270, 300 degrees, (b) named error = —
4. (a) all solutions = 0, 270 degrees, (b) named error = —
5. (a) all solutions = 15, 75, 195, 255 degrees, (b) named error = —
6. (a) correct simplification = $1 + 2\cos(\theta)$, (b) value at 60 degrees = 2, (c) named error = —
7. (a) all solutions = 0, 120, 240 degrees, (b) named error = —
8. (a) all solutions = 0, 30, 150, 180, 210, 330 degrees, (b) both errors named = —

What is verified

9 of 17 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus
HSF-TF.C.8 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 17 tagged checks.

Common wrong answers

1. *If they got 60, 300: divided both sides by sine, which throws away every angle where the sine is zero*
2. *If they got 30, 150: took only the positive square root, dropping the whole negative case*
3. *If they got 60, 300: cancelled a cosine from both sides, losing the two angles at which the cosine is zero*
4. *If they got 0, 180, 270: squared both sides and reported every root of the squared equation without testing them in the original*
5. *If they got 15, 75: solved the doubled angle over the original interval instead of the doubled one, so half the solutions were never found*
6. *If they got 3: cancelled the sine out of the double-angle term as well and lost the cosine factor with it*
7. *If they got 120, 240: decided the factor giving a cosine of one had no solution and dropped it*

8. *If they got 30, 150: cancelled the sine factor and also solved the doubled angle over the original interval, losing solutions twice over*

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1. $2\sin\theta\cos\theta = \sin\theta$ on $0° \le \theta < 360°$. Student divided by $\sin\theta$ and got $60°, 300°$.

(a) Move everything to one side and factor — never divide by something that can be zero.

$$2\sin\theta\cos\theta - \sin\theta = 0 \quad\Longrightarrow\quad \sin\theta\,(2\cos\theta - 1) = 0$$

$\sin\theta = 0$ gives $\theta = 0°, 180°$; $\cos\theta = \dfrac{1}{2}$ gives $\theta = 60°, 300°$. $\boxed{\theta = 0°,\ 60°,\ 180°,\ 300°}$

(b) *Instructor-judged.* The response must say that dividing by $\sin\theta$ is legal only where $\sin\theta \neq 0$, and that the angles where it *is* zero are precisely the two solutions destroyed. Full credit also states the repair: collect on one side and factor, so the sine survives as a factor. Noting only that two answers are missing is partial credit.

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2. $4\sin^2\theta - 1 = 0$ on $0° \le \theta < 360°$. Student took one square root and got $30°, 150°$.

(a) Factor as a difference of squares, which cannot lose a branch:

$$(2\sin\theta - 1)(2\sin\theta + 1) = 0 \quad\Longrightarrow\quad \sin\theta = \tfrac{1}{2} \ \text{ or } \ \sin\theta = -\tfrac{1}{2}.$$

The positive case gives $30°, 150°$; the negative case gives $210°, 330°$.

$\boxed{\theta = 30°,\ 150°,\ 210°,\ 330°}$

(b) *Instructor-judged.* The response must say a square equal to a positive number has *two* square roots, so $\sin\theta = \pm\dfrac{1}{2}$, and only the positive branch was kept. Each branch supplies two angles per turn, so four solutions exist. Recommending the difference-of-squares factorisation earns full credit.

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3. $2\cos^2\theta = \cos\theta$ on $0° \le \theta < 360°$. Student divided by $\cos\theta$ and got $60°, 300°$.

(a)

$$2\cos^2\theta - \cos\theta = 0 \quad\Longrightarrow\quad \cos\theta\,(2\cos\theta - 1) = 0$$

$\cos\theta = 0$ gives $\theta = 90°, 270°$; $\cos\theta = \dfrac{1}{2}$ gives $\theta = 60°, 300°$. $\boxed{\theta = 60°,\ 90°,\ 270°,\ 300°}$

(b) *Instructor-judged.* The response must identify the cancelled cosine and say that cancelling silently assumes $\cos\theta \neq 0$ — an assumption the problem never grants. Factoring keeps both cases. Full credit requires naming the assumption, not just listing the two missing right angles.

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4. $\sin\theta + 1 = \cos\theta$ on $0° \le \theta < 360°$. Student squared and reported $0°, 180°, 270°$.

(a) The squared equation does give those three candidates, but squaring is not reversible, so each must be tested in the *original* equation:

$$\begin{array}{rlr} \theta = 0°: & 0 + 1 = 1 \ \text{ and } \ \cos 0° = 1 & \checkmark \\ \theta = 180°: & 0 + 1 = 1 \ \text{ but } \ \cos 180° = -1 & \times \\ \theta = 270°: & -1 + 1 = 0 \ \text{ and } \ \cos 270° = 0 & \checkmark \end{array}$$

$$\boxed{\theta = 0^\circ,\ 270^\circ}$$

(b) *Instructor-judged.* The response must say squaring can create roots the original never had, so every candidate has to be substituted back, and must carry out that test on 180° to show the two sides disagree. Saying only that one answer is wrong, without the reason squaring produces it, is partial credit.

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5. $2\sin 2\theta = 1$ on $0^\circ \le \theta < 360^\circ$. Student reported $15^\circ, 75^\circ$.

(a) Put $u = 2\theta$. As θ runs over $[0^\circ, 360^\circ)$, u runs over $[0^\circ, 720^\circ)$:

$$\sin u = \tfrac{1}{2} \quad \Longrightarrow \quad u = 30^\circ,\ 150^\circ,\ 390^\circ,\ 510^\circ.$$

Halving each: $\theta = 15^\circ, 75^\circ, 195^\circ, 255^\circ$. $\boxed{\theta = 15^\circ,\ 75^\circ,\ 195^\circ,\ 255^\circ}$

(b) *Instructor-judged.* The response must say the doubled angle sweeps twice as far, so its interval ends at 720°, and stopping at 360° discards the second pass. Full credit states the repair: substitute for the inside angle, multiply both endpoints by the coefficient, solve, then divide every answer back.

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6. Simplify $\dfrac{\sin\theta + \sin 2\theta}{\sin\theta}$. Student cancelled every $\sin\theta$ and wrote 3.

(a) Expand the double angle first, then factor the numerator before dividing:

$$\frac{\sin\theta + 2\sin\theta\cos\theta}{\sin\theta} = \frac{\sin\theta\,(1 + 2\cos\theta)}{\sin\theta} = 1 + 2\cos\theta \quad (\sin\theta \neq 0).$$

$$\boxed{1 + 2\cos\theta}$$

(b) At $\theta = \dfrac{\pi}{3}$, $\cos\theta = \dfrac{1}{2}$, so the expression is $1 + 2\left(\dfrac{1}{2}\right)$. $\boxed{2}$

(c) *Instructor-judged.* The response must say a common factor may only be cancelled from a whole term: the second term is $2\sin\theta\cos\theta$, so removing its $\sin\theta$ leaves $2\cos\theta$, not a bare 2. Full credit requires naming the dropped cosine factor; a correct answer shows the numerator written as $\sin\theta(1 + 2\cos\theta)$ before dividing.

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7. $\cos 2\theta = \cos\theta$ on $0^\circ \le \theta < 360^\circ$. Student reported $120^\circ, 240^\circ$ and discarded $\cos\theta = 1$.

(a) The factoring is correct: $(2\cos\theta + 1)(\cos\theta - 1) = 0$. Both factors matter.

$$\cos\theta = -\tfrac{1}{2} \implies \theta = 120^\circ,\ 240^\circ; \qquad \cos\theta = 1 \implies \theta = 0^\circ.$$

$$\boxed{\theta = 0^\circ,\ 120^\circ,\ 240^\circ}$$

(b) *Instructor-judged.* The response must say that $\cos\theta = 1$ is perfectly attainable — it happens at 0°, which the half-open interval includes — so that factor contributes a solution. Full credit notes that 360° is excluded, which is why the angle appears once rather than twice. Supplying the missing angle without saying why the student wrongly rejected it is partial credit.

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8. $2 \sin\theta \cos 2\theta = \sin\theta$ on $0° \le \theta < 360°$. Student cancelled $\sin\theta$, used $2\theta \in [0°, 360°)$, and reported $30°, 150°$.

(a) Factor rather than cancel, then widen the interval for the doubled angle:

$$\sin\theta\,(2\cos 2\theta - 1) = 0$$

$\sin\theta = 0$ gives $\theta = 0°, 180°$. For the other factor, $u = 2\theta$ runs over $[0°, 720°)$ and $\cos u = \dfrac{1}{2}$ at $u = 60°, 300°, 420°, 660°$, so $\theta = 30°, 150°, 210°, 330°$.

$$\boxed{\theta = 0°,\ 30°,\ 150°,\ 180°,\ 210°,\ 330°}$$

(b) *Instructor-judged.* Two independent faults must be named, with the losses attributed. Dividing by $\sin\theta$ cost $0°$ and $180°$. Solving the doubled angle over the un-widened interval cost $210°$ and $330°$. Naming one fault only is half credit; a correct response says which missing solutions came from which fault.